

Yanfei Zhu

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EDUCATION

University of Wisconsin-Madison

Madison, WI

M.A. in Mathematics, GPA: 3.844/4.000

Sep. 2024 – May 2026

Visiting International Student Program (VISP)

- **Courses:** Introductory Topology I&II (Algebraic Topology), Homological Algebra, Abstract Algebra, Modern Algebra
- Began UW coursework during final year of undergraduate studies; returned to SUSTech for B.S. defense; resumed UW to complete M.A.

Southern University of Science and Technology (SUSTech)

Shenzhen, CN

B.S. in Mathematics and Applied Mathematics

Sep. 2021 – Jun. 2025

- **Weighted Average Score:** 88.09/100
- **Courses:** Algebraic Topology, Topology, Representation of Groups, Abstract Algebra, Advanced Linear Algebra, Discrete Mathematics, Elementary Number Theory

Research Interests: I am interested in using algebraic and topological methods to study geometric problems, especially those involving manifolds. My current research focuses on equivariant homotopy theory and homological stability. I am also interested in optimization on manifolds.

RESEARCH AND READING EXPERIENCE

Directed Reading in Manifold Optimization

Jun. 2026 – Present

Advisor: Prof. Qinglin Tang, Sichuan University

Topics: Manifold optimization

Directed Reading in Intersection Homology

Aug. 2025 – Dec. 2025

Advisor: Prof. Laurentiu Maxim, Mathematics Department, UW–Madison

Topics: Intersection homology, sheaves on topological spaces, Verdier duality

Directed Reading in Riemannian Geometry and Morse Theory

Mar. 2025 – Oct. 2025

Advisor: Prof. Ruobing Zhang, Mathematics Department, UC San Diego

Topics: Riemannian geometry (Petersen, do Carmo), Morse theory (Milnor)

Directed Reading in Homological Stability

Apr. 2025 – Jul. 2025

Advisor: Prof. Yifei Zhu, Mathematics Department, SUSTech

Topics: Homological stability, configuration spaces, FI-modules, braid groups

Undergraduate Thesis

Dec. 2024 – Jul. 2025

Symmetry of Solutions in Nonlinear Aggregation-Diffusion Equations and Its Applications

Advisor: Prof. Linlin Su, SUSTech

Studied symmetry properties of stationary solutions to nonlinear aggregation-diffusion equations, with a focus on radial symmetry and applications to numerical methods.

Reading Seminar in Homological Algebra

Sep. 2023 – Dec. 2023

Advisor: Prof. Ziqing Xiang, Mathematics Department, SUSTech

Topics: Category theory, homological algebra

Preprints

The $RO(C_3)$ -graded cohomology of a complete flag variety

Preprint, 2026.

Computes the genuine equivariant cohomology ring, Schubert products, and fixed-point restriction.

Two Lifted Actions on Oriented Configuration Spaces: Stabilization and Bredon Homology

Preprint, 2026.

Finite generation of integral Bredon homology for oriented configuration spaces, using both lifts of a reflection action.

TEACHING EXPERIENCE

Grader, Department of Mathematics, UW–Madison

Feb. 2026 – May 2026

Graded undergraduate mathematics assignments and provided written feedback.

SUMMER SCHOOLS AND WORKSHOPS

Summer School, University of Science and Technology of China

Jul. 2025

Hefei, CN. Topics: Geometry and analysis.

IWoAT Summer School, Westlake University

Jul. 2025

Hangzhou, CN. Topics: Adams spectral sequence computations; the last Kervaire invariant problem.

SKILLS AND LANGUAGES

Technical Skills: LaTeX, Python, MATLAB, Java

Languages: Mandarin (native), English (fluent), Japanese (basic)